Leveraging G×E interaction to optimize multi-trait selection in Lima Bean

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```r
load("../Data/ebvMat.rda")
```

0.1 Desire index

```r
# DESIRE INDEX FUNCTION

gainDiff <- function(ebvMat, b_ebv, dGain, nSel){

  take <- order(ebvMat %*% b_ebv, decreasing = TRUE)[1:nSel]

  mu_Sel <- colMeans(ebvMat[take, ])
  resp   <- mu_Sel - colMeans(ebvMat)
  resp   <- resp / sqrt(sum(resp^2))

  sum((resp - dGain)^2)
}
```

```r
ebvMat <- ebvMat[complete.cases(ebvMat), ]
#COV MATRIX -EBVS
library(AlphaSimR)
```

```
## Loading required package: R6
```

```r
G_ebv <- AlphaSimR::popVar(ebvMat)
G_ebv
```

```
##            [,1]       [,2]       [,3]       [,4]       [,5]       [,6]
## [1,] 6587.97833 217.258861 424.765720 256.943342 36.6766871 4080.22632
## [2,]  217.25886   7.323476  14.181373   8.521450  1.2173849  134.59974
## [3,]  424.76572  14.181373  28.804036  17.134163  2.4397036  266.63926
## [4,]  256.94334   8.521450  17.134163  10.367730  1.4703171  160.61068
## [5,]   36.67669   1.217385   2.439704   1.470317  0.2129143   22.92692
```

```
## [6,] 4080.22632 134.599743 266.639262 160.610683 22.9269219 2564.70381
```

```r
# G_ebv |>  kableExtra::kable() |> kableExtra::kable_styling()
```

```r
#desired gains
desiredGain <- c(
  GY   =  2,
  OHSW =  1,
  NDF  = -1,
  NDM  = -1,
  PH   =  0,
  PN   =  1
)
desiredGain <- desiredGain / sqrt(sum(desiredGain^2))
sum(desiredGain^2)
```

```
## [1] 1
```

```r
b_ebv <- solve(G_ebv) %*% desiredGain
b_ebv
```

```
##             [,1]
## [1,] -0.07014002
## [2,]  2.70032556
## [3,] -0.38420756
## [4,] -2.08437923
## [5,]  8.33887739
## [6,]  0.06593790
```

```r
nSel <- 5
opt <- optim(
  par     = b_ebv,
  fn      = gainDiff,
  ebvMat  = ebvMat,
```

```r
  dGain   = desiredGain,
  nSel    = nSel
)
opt_weights <- opt$par
opt_weights
```

```
##             [,1]
## [1,] -0.06570972
## [2,]  2.70035237
## [3,] -0.38474376
## [4,] -2.08393017
## [5,]  8.33759052
## [6,]  0.06765372
```

```r
DesireIndex <- ebvMat %*% opt_weights

DesireIndex <- data.frame(
  GEN = rownames(ebvMat),
  DesireIndex = as.numeric(DesireIndex)
) |>
  dplyr::arrange(desc(DesireIndex))

head(DesireIndex, 10)
```

```
##    GEN DesireIndex
## 1  L06    234.5246
## 2  L27    234.2351
## 3  L21    233.6062
## 4  L20    233.5220
## 5  L13    233.4020
## 6  L34    233.3338
```

```
## 7  L14    232.8109
## 8  L11    232.7500
## 9  L25    232.4086
## 10 L12    232.2843
```

```r
selected <- DesireIndex[1:5, ]
selected
```

```
##   GEN DesireIndex
## 1 L06    234.5246
## 2 L27    234.2351
## 3 L21    233.6062
## 4 L20    233.5220
## 5 L13    233.4020
```